

COMPARATIVE STUDY OF POLY(ETHYLENE OXIDE) ELECTROLYTES MADE WITH $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, LiCF_3SO_3 AND LiClO_4 : THERMAL PROPERTIES AND CONDUCTIVITY BEHAVIOUR

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Abstract—The phase transitions in mixtures of poly(ethylene oxide) (PEO) ($M_n = 4 \times 10^3$) with $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, a lithium salt of low lattice energy recently discovered by Armand and coworkers, are compared with those of the PEO– LiCF_3SO_3 and PEO– LiClO_4 systems. As expected, this new salt is more soluble in PEO than LiCF_3SO_3 and LiClO_4 . It forms crystalline complexes similar to those of LiClO_4 , but its mixtures made in 10/1 and 8/1 molar ratios (EO units/salt) remained completely amorphous, even after a long standing at -5°C . In agreement with Armand and coworkers, these electrolytes exhibit conductivity magnitudes greater than $10^{-5} \text{ S cm}^{-1}$ at 25°C . A comparison made at higher temperatures (50 and 100°C) with melted or supercooled electrolytes containing LiClO_4 in the same molar ratios is slightly in favour of $\text{LiN}(\text{CF}_3\text{SO}_2)_2$. The solubility curve of LiClO_4 has been characterized over a wide range of temperatures. A thermodynamical analysis of this curve yields negative values for both the enthalpy and entropy of mixing of the liquid components.

Key words: poly(ethylene oxide), $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, LiCF_3SO_3 , LiClO_4 , phase diagrams.

INTRODUCTION

Cation solvation in amorphous polyethers leads to a peculiar situation where the presence of ions in a liquid-like electrolyte has a dramatic effect upon both the kinetic energy and the entropy of the system. This effect arises because the cation–oxygen binding energy, which is the driving force for salt dissolution, contributes to the increase in the barrier to rotation of the polymer segments. This is evidenced by a marked increase in glass transition temperature (T_g) with increasing salt content. Although many questions related to the dynamics of polyether–salt amorphous complexes have been clarified[1], little is known concerning the thermodynamics of the solvation process in these materials.

In previous works[2, 3], solubility data derived from phase diagram construction and T_g measurements were reported for a series of alkali thiocyanates complexed with poly(ethylene oxide) (PEO) and poly(methyl glycidyl ether) (PMGE). From these data, evidence was provided that solvates of well-defined stoichiometries form in both polymers. Furthermore, the comparative study of PEO and PMGE allowed a semi-quantitative interpretation of the solvate stability in terms of binding energy and configurational entropy[3]. A similar investigation based upon thermal properties has been extended to a series of lithium salts including $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, a salt of low lattice energy recently discovered by Armand *et al.*[4]. According to these authors, the large, flexible anion of this salt acts as a plasticizer in its complexes with PEO. Indeed, they reported the preparation of PEO– $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ complexes of both low crystallinity and low T_g that exhibited

conductivity magnitudes greater than $10^{-5} \text{ S cm}^{-1}$ at 25°C [4].

In the present paper, a comparison is made of the phase diagrams of the PEO– $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, PEO– LiCF_3SO_3 , and PEO– LiClO_4 systems. Also compared, are the T_g –composition relationships of supercooled mixtures of these systems, and conductivity isotherms (at 50 and 100°C) of $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ and LiClO_4 over a range of compositions in the concentrated regime. The solubility data obtained for LiClO_4 cover a wide range of temperatures. They are analysed in terms of the Flory–Huggins concepts which allow the definition of the enthalpy and entropy of mixing of the liquid components. All the data were obtained with a low molecular weight PEO sample having a degree of crystallinity of 96%[5].

EXPERIMENTAL

The detail concerning the purification and the characterization of the PEO sample ($M_n = 3.9 \times 10^3$, $M_w/M_n = 1.02$) was reported in a previous work[2]. LiCF_3SO_3 (Aldrich, 97%) and LiClO_4 (K&K, 99.8%) were dried under high vacuum at 130°C . $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ (kindly supplied by M. Gauthier of IREQ) was dried at 160°C . This higher temperature was necessary to achieve a complete dehydration of this salt. PEO–salt mixtures were prepared under a dry atmosphere by mixing weighted quantities of 1–5% methanol solutions of each component. Solvent evaporation was carried out in ampoules connected to a vacuum system. The solvent-free mixtures were subsequently dried under high vacuum at either 130°C (LiCF_3SO_3 and LiClO_4) or 160°C

($\text{LiN}(\text{CF}_3\text{SO}_2)_2$) and allowed to crystallize for many months at a temperature *ca* 20°C.

Melting (or dissolution) endotherms and glass transition anomalies were recorded at heating rates of 10 and 40°C min⁻¹, respectively, with a Perkin-Elmer DSC-4 calorimeter. Supercooled specimens were obtained by melt quenching at a cooling rate of 320°C min⁻¹. The values of T_g were read at the intersection of the tangent drawn through the heat capacity jump with the base line recorded before the transition. Conductivity measurements were carried out on amorphous electrolytes only. The samples were heated to 130–160°C prior to the measurements. A Hewlett-Packard 4192A impedance analyser was used together with cells similar to that described in a previous work[2].

RESULTS AND DISCUSSION

Figures 1–3 summarize the thermal transitions recorded by DSC upon the first heating of a series of mixtures prepared with each salt. In these phase diagrams, the vertical boundaries, which characterize the stoichiometry of the intermediate crystalline compounds, were derived from calorimetric analysis of the DSC data. They correspond to the compositions at which the heat of fusion (or dissolution) related to each compound exhibited a maximum value. Due to the complexity of the phase diagram of the PEO–LiClO₄ system in the salt-rich region, X-ray diffraction data were used to confirm the stoichiometries of the ultimate compounds of this system.

Most of the features in the phase diagrams of the PEO–LiCF₃SO₃ and PEO–LiClO₄ systems are in agreement with anterior data reported on these systems. An account of these previous works was given

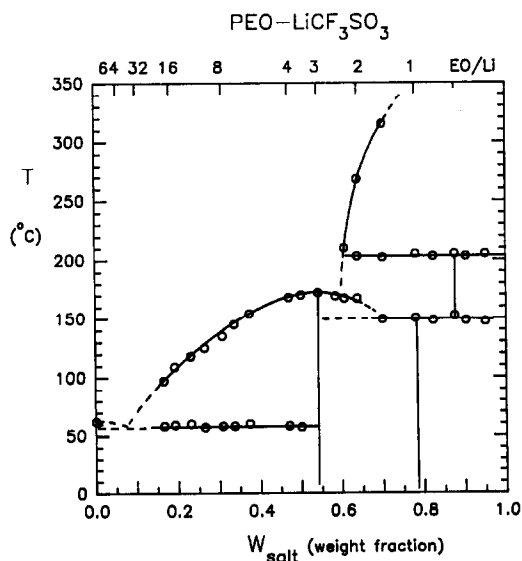


Fig. 1. Phase diagram of the PEO–LiCF₃SO₃ system. The vertical boundaries, which indicate the formation of 3/1, 1/1, and 0.5/1 crystalline compounds, were derived from calorimetric analysis of the DSC data. The 0.5/1 compound forms upon the incongruent melting of the 1/1 compound at 150°C.

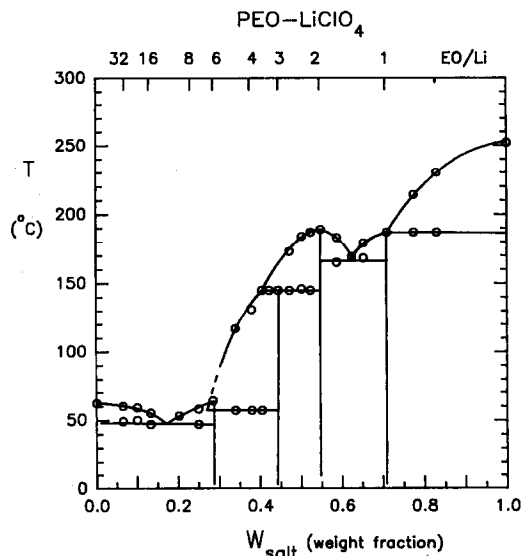


Fig. 2. Phase diagram of the PEO–LiClO₄ system. The vertical boundaries, which indicate the formation of 6/1, 3/1, 2/1, and 1/1 crystalline compounds, were derived from both calorimetric analysis and X-ray diffraction data.

in a recent review by Fauteux[6]. Therefore, the discussion of these systems will be limited to the salt-rich region which was either not covered in these former works, or covered over a partial range of temperatures, as in a recent work reported by Zahurak *et al.*[7] on the PEO–LiCF₃SO₃ system.

In Fig. 1, it may be seen that a discontinuity occurs near the 2/1 molar ratio (EO units/salt) in the phase

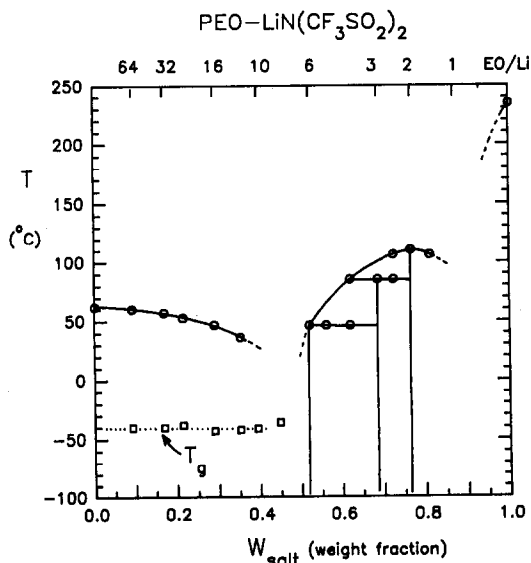


Fig. 3. Partial phase diagram of the PEO–LiN(CF₃SO₂)₂ system. The vertical boundaries, which indicate the formation of 6/1, 3/1, and 2/1 crystalline compounds, were derived from calorimetric analysis of the DSC data. The mixtures with molar ratios greater than 6/1 were either amorphous (EO/Li = 8 and 10) or semicrystalline (EO/Li > 10). The T_g values of the amorphous phase in the latter mixtures (dotted line) correspond to that of the 10/1 mixture.

diagram of the PEO–LiCF₃SO₃ system. This discontinuity is due to the fact that the 1/1 compound undergoes a peritectic reaction at a temperature which is lower than the melting point of the 3/1 compound. The important point, however, concerns the liquidus curve of LiCF₃SO₃. This curve is obtained above the tie line at 203°C which is related to the incongruent melting of the 0.5/1 compound. It may be seen that the solubility of LiCF₃SO₃, which corresponds to a weight fraction $W_{\text{salt}} = 0.61$ (2.3/1 molar ratio) at 203°C, increases with increasing temperature above this limit. Unfortunately, because the mixtures were not stable at temperatures above 300°C, it has been impossible to characterize the solubility curve up to the melting point of the salt. Also note that LiCF₃SO₃ undergoes a thermal decomposition at its melting point *ca* 430°C.

Fortunately, this is not the case for the other salts. The melting point of LiClO₄ is 251°C, while that of LiN(CF₃SO₂)₂ is only 234°C. In Fig. 2, the liquidus curve of LiClO₄ is obtained above the tie line at 187°C which is related to the incongruent melting of the 1/1 compound. It may be seen that the solubility of this salt at 203°C corresponds to a weight fraction $W_{\text{salt}} = 0.74$ (0.85/1 molar ratio). It is greater than that of LiCF₃SO₃ at the same temperature. Furthermore, it is clear that LiClO₄ dissolves in all proportions in PEO at elevated temperatures. This feature apparently applies to LiN(CF₃SO₂)₂, though the phase diagram of the corresponding system (Fig. 3) has not yet been completed beyond $W_{\text{salt}} = 0.81$ (1.5/1 molar ratio). Like LiClO₄, LiN(CF₃SO₂)₂ forms 6/1, 3/1, and 2/1 crystalline compounds with PEO. The difference, however, is that these compounds melt at considerably lower temperatures than those related to LiClO₄. This difference was not due to partial crystallinity since, over this range of compositions, the DSC curves of all the mixtures including the 6/1 mixture did not exhibit any T_g anomaly.

Another feature of the PEO–LiN(CF₃SO₂)₂ system concerns its mixtures having molar ratios greater than 6/1. As it may be seen in Fig. 3, none of these mixtures exhibited thermal events (eutectic or dissolution endotherms) related to the 6/1 compound. Moreover, the mixtures with 10/1 and 8/1 molar ratios remained completely amorphous, even after a standing of two months at –5°C in a freezer. Note that Armand *et al.*[4] reported DSC curves for similar mixtures prepared with a high molecular weight PEO that exhibited strong T_g anomalies near –40°C together with small endotherms at 53°C. With the present PEO, crystallinity was observed only for the mixtures with molar ratios greater than 10/1. A plot as a function of W_{salt} of the heat of dissolution per gram of sample (ΔQ) corresponding to the endotherms recorded for these mixtures yielded a linear curve with a ΔQ -axis intercept of 185 J g^{–1} ($\Delta H_f = 192$ J g^{–1} for the present PEO), and a composition-axis intercept of 0.37 (11/1 molar ratio). This feature, together with the invariance of T_g (–40 ± 3°C) for the amorphous phases in these semicrystalline mixtures, is probably due to the fact that they were all equilibrated for many months at 20°C prior to their study. Therefore the possibility that a eutectic crystallization could take place at a lower temperature remains to be clarified.

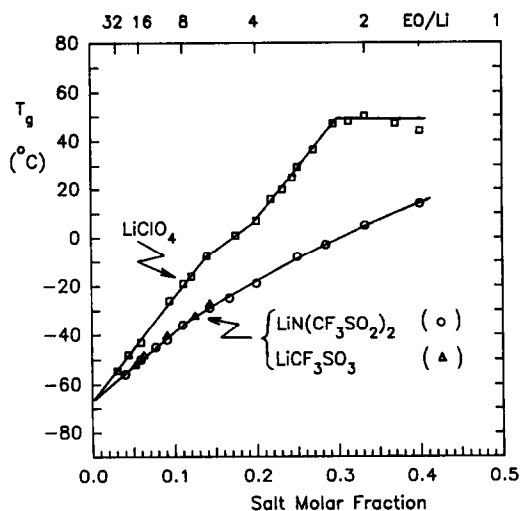


Fig. 4. Comparison of the T_g -composition relationships obtained for supercooled mixtures of the PEO–LiClO₄, PEO–LiCF₃SO₃, and PEO–LiN(CF₃SO₂)₂ systems. The data of the PEO–LiCF₃SO₃ system are limited to molar ratios greater than 6/1 because the mixtures more concentrated in LiCF₃SO₃ could not be supercooled by melt quenching.

Figure 4 shows the T_g -composition relationships obtained for supercooled mixtures of the three systems. The comparison is made on a molar basis and, except for LiCF₃SO₃, it extends over a range of molar ratios decreasing from *ca* 32/1 to 1.5/1. In the case of LiCF₃SO₃, the data are limited to molar ratios greater than 6/1 because more concentrated mixtures could not be supercooled by melt quenching. Over this partial range, the T_g elevation produced by LiN(CF₃SO₂)₂ appears to be identical with that produced by LiCF₃SO₃ (2.8°C/mol%). It is markedly inferior to that produced by LiClO₄ (4.3°C/mol%). For salt contents just above this range, a change of regime is observed for the PEO–LiClO₄ system that suggests a structural rearrangement. Beyond this region, the T_g -composition relationship of this system again shows a linear rise up to a ceiling value of 49°C. From this feature, it may be inferred that the solubility of LiClO₄ at 49°C corresponds to a 2.3/1 molar ratio ($W_{\text{salt}} = 0.51$). LiN(CF₃SO₂)₂ appears to be considerably more soluble than LiClO₄ since, over the same range of compositions, there is no evidence of saturation in the T_g -composition curve of this salt. This curve rather exhibits a decelerated rise with increasing salt content. This feature apparently confirms that the large, flexible anion of this salt acts as a plasticizer[4].

The liquidus curve of LiClO₄ in Fig. 2 is the analogue of the melting point depression of this salt by the diluent PEO. Note that the foregoing data concerning the solubility of LiClO₄ at 49°C also belong to this liquidus curve. Therefore, advantage was taken of this feature to proceed to a thermodynamical analysis based upon the Flory–Huggins concepts[8]. According to these concepts, the melting point depression of the salt (denoted by the subscript s) by the polymeric diluent (denoted by the subscript p) should be governed by the magnitude of the

Flory-Huggins interaction parameter χ according to the relation

$$\frac{1}{T_f} - \frac{1}{T_f^0} = -\frac{R}{\Delta H_{f,s}} \left[\ln(1 - \phi_p) + \left(1 - \frac{1}{x}\right) \phi_p + \chi \phi_p^2 \right], \quad (1)$$

where $\Delta H_{f,s}$ is the molar heat of fusion of the salt, ϕ_p is the volume fraction of the polymer, x is the number of segments per chain having the same partial molar volume V_s as that of the salt, and R is the perfect gas constant.

The parameter χ in equation (1) is a measure of the excess free enthalpy of mixing, $\Delta G_{m,e}$, of the liquid components[8]. When this latter quantity is written per unit volume of mixture (denoted by the superscript v), it is given by the relation

$$(\Delta G_{m,e})^v = RT\chi\phi_p(1 - \phi_p)/V_s. \quad (2)$$

By assuming that χ is composition independent and changes with temperature according to the simple relation

$$\chi = (a/T) + b, \quad (3)$$

equation (2) can be divided into enthalpic and entropic terms:

$$(\Delta H_{m,e})^v = Ra\phi_p(1 - \phi_p)/V_s, \quad (4)$$

$$(\Delta S_{m,e})^v = -Rb\phi_p(1 - \phi_p)/V_s. \quad (5)$$

Note that, according to this formalism[8], the overall enthalpy, entropy, and free enthalpy of mixing of the liquid components are respectively

$$\Delta H_m^v \equiv (\Delta H_{m,e})^v \quad (6)$$

$$\Delta S_m^v = (\Delta S_{m,e})^v - (R/V_s)[(1 - \phi_p)\ln(1 - \phi_p) + (\phi_p/x)\ln\phi_p] \quad (7)$$

$$\Delta G_m^v = \Delta H_m^v - T\Delta S_m^v, \quad (8)$$

where the term with square brackets in equation (7) is the ideal, combinatorial/configurational contribution to the entropy of mixing.

Figure 5 shows a plot as a function of $1/T$ of the values of χ obtained by applying equation (1) to the solubility data of LiClO_4 . The values of ϕ_p were estimated from the weight fractions of PEO by assuming that the density ratio $\rho(\text{LiClO}_4)/\rho(\text{PEO})$ is temperature independent and close to that of the crystalline components at 25°C, that is, $\rho(\text{LiClO}_4) = 2.43 \text{ g cm}^{-3}$ and $\rho(\text{PEO}) = 1.23 \text{ g cm}^{-3}$. The value of x was estimated to be 72 as computed from the same density ratio, and the value of $\Delta H_f(\text{LiClO}_4)$ was 14.6 kJ mol^{-1} as measured by DSC. In agreement with equation (3), the plot of χ as a function of $1/T$ yields a linear relationship with a slope $a = -2.56 \times 10^3 \text{ K}$, and an intercept $b = 4.00$. According to these results, the value of χ should remain negative (*ie* favorable to miscibility) up to 367°C, that is, up to temperatures well above the melting point of LiClO_4 .

When the foregoing values of a and b are used in equations (4) and (5) to compute the excess enthalpy and the excess entropy of mixing for a hypothetical mixture containing 50% by volume of PEO ($W_{\text{salt}} = 0.66$), one obtains the values $(\Delta H_{m,e})^v = -122 \text{ J cm}^{-3}$ (or $-10.7 \text{ kJ mol}^{-1}$ of salt) and $(\Delta S_{m,e})^v = -0.190 \text{ J K}^{-1} \text{ cm}^{-3}$ (or $-16.6 \text{ J K}^{-1} \text{ mol}^{-1}$ of salt). These data

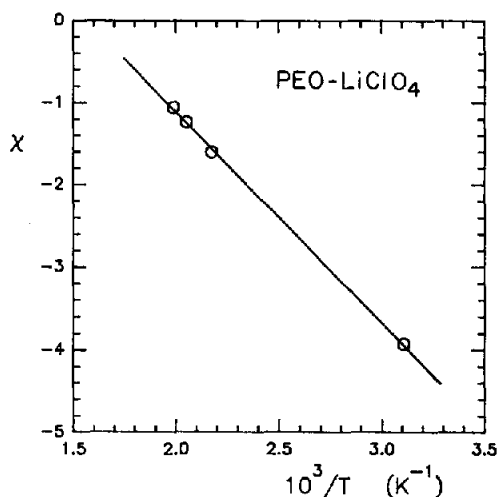


Fig. 5. Plot as a function of $1/T$ of the Flory-Huggins interaction parameter χ derived from the solubility data of LiClO_4 . The three upper data points correspond to the solubility data depicted in the phase diagram of the PEO-LiClO_4 system (Fig. 2), while the lowest data point corresponds to the solubility (at 49°C) derived from the T_g -composition relationship in Fig. 4.

(computed with $V_s = 43.8 \text{ cm}^3 \text{ mol}^{-1}$) show that a small but significant cost in entropy results from the solvation of the ions. However, even at the melting point of the salt, this cost remains substantially inferior to the energy liberated by the reaction. Indeed, at this temperature, the value of ΔG_m^v computed by

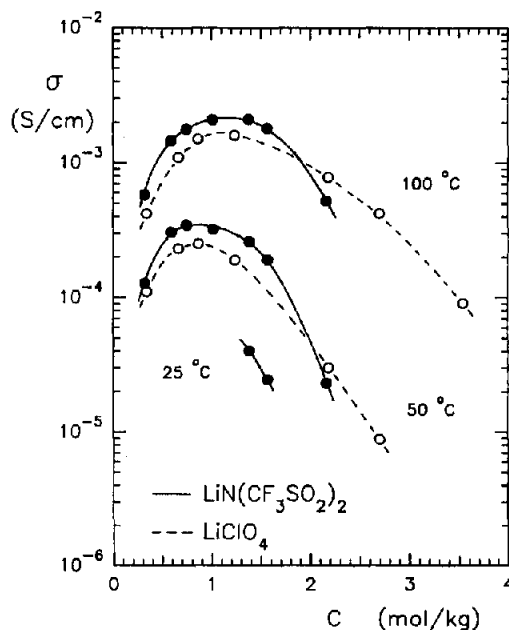


Fig. 6. Plots as a function of the salt concentration (in mol kg^{-1} of mixture) of the conductivity σ measured on melted or supercooled mixtures of $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ and LiClO_4 at 50 and 100°C, respectively. The data at 100°C cover a range of molar ratios extended from 64/1 to 4/1 for both salts. Also included are the conductivity data obtained at 25°C for the 10/1 and 8/1 amorphous mixtures prepared with $\text{LiN}(\text{CF}_3\text{SO}_2)_2$.

means of equations (6) to (8) for a 50% by volume mixture is -57 J cm^{-3} (or -5.0 kJ mol^{-1} of salt).

In Fig. 6 a comparison is made at two temperatures (50 and 100°C) of the conductivity data obtained for $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ and LiClO_4 . These data, which correspond to melted or supercooled electrolytes, are plotted as a function of the salt concentration (c) in mol kg^{-1} of mixture. At 100°C , they cover a range of molar ratios from 64/1 to 4/1 for both salts. It may be seen that $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ is slightly more conductive than LiClO_4 over the range of concentrations, $0.6 < c < 1.4 \text{ mol kg}^{-1}$, where optimal results are obtained for both systems. Conductivity data measured at 25°C on the amorphous mixtures containing $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ in 10/1 and 8/1 molar ratios are also depicted in Fig. 6. These data, as well as the foregoing data related to this salt, are comparable to those reported by Armand *et al.*[4] for similar electrolytes made with a high molecular weight PEO sample. At 25°C , the mixtures with 10/1 and 8/1 molar ratios exhibit conductivities of 4×10^{-5} and $2.5 \times 10^{-5} \text{ S cm}^{-1}$, respectively. Therefore, if these mixtures, as well as mixtures of the same compositions prepared with high molecular weight PEO, were proven to remain highly amorphous under more severe conditions, $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ would appear to be an outstanding salt for granting room-temperature operation of PEO-based lithium batteries.

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